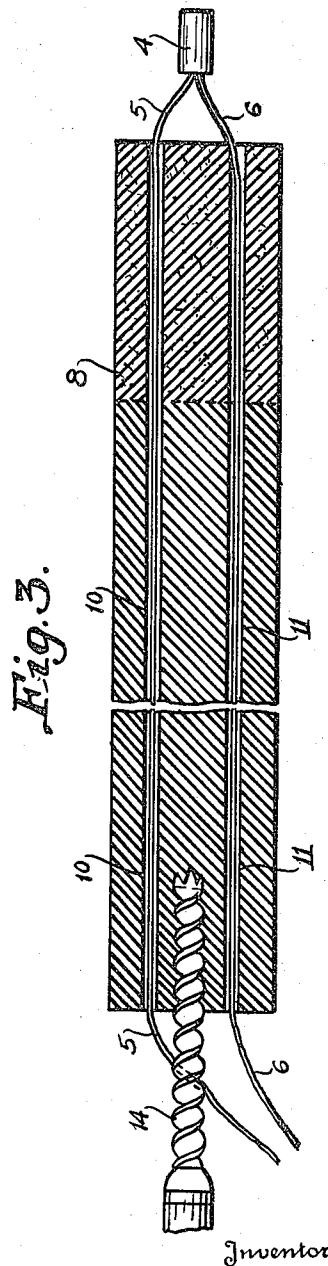
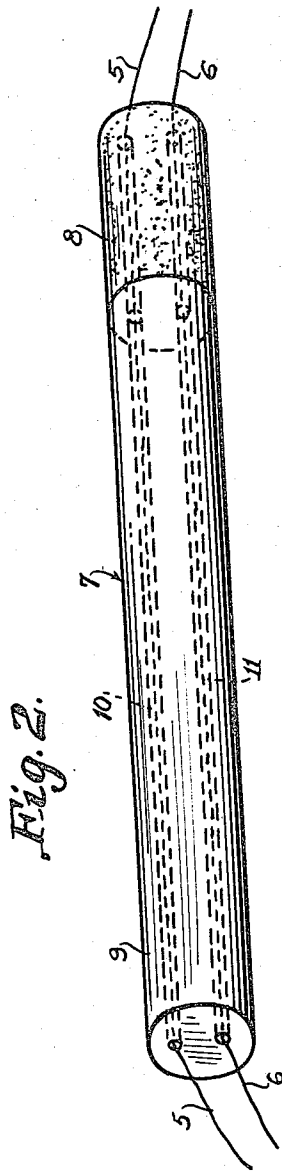
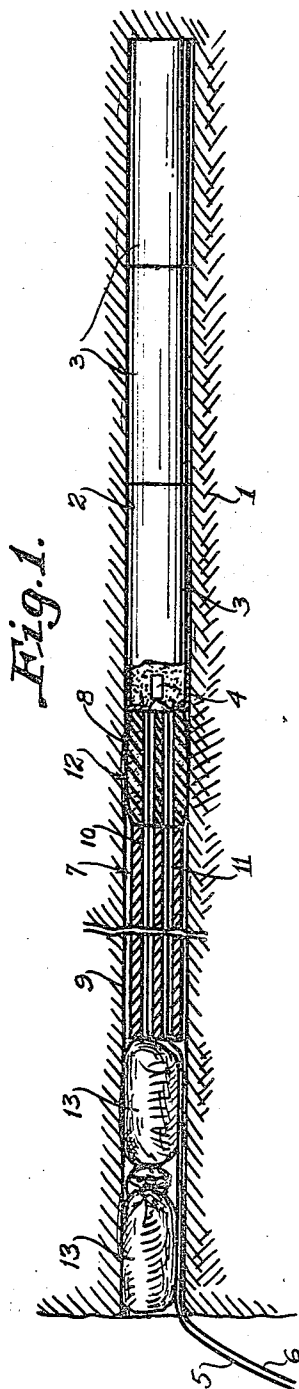


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R. J. HEITZMAN
SAFETY BLASTING PLUG
Filed Aug. 18, 1932

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UNITED STATES PATENT OFFICE

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SAFETY BLASTING PLUG

Application filed August 18, 1932. Serial No. 629,347.

This invention relates to the promotion of safety in blasting, as employed particularly in mining and quarrying, and it has for its object the provision of a safety blasting plug which, by its structure, and the incident simplicity in handling and installing, avoids those causes of misfire ordinarily attributable to short circuiting, and should a misfire occur through inherent defect in the detonating cap, the invention affords a safe means for the extraction of the detonating cap from the bore hole.

More specifically, the invention has for its object the provision of a safety blasting plug characterized by the entire absence of metal in its construction, and constituted by an integral cylindrical body, the advance ends of which are expansible radially under pressure to conform to the wall of the bore hole in a firm and gas-tight manner, the rear portion being of relatively rigid material capable of being penetrated by an auger for the purpose of effecting the withdrawal of the plug.

Another object of the invention is the provision of a blasting plug comprising a unitary non-metallic cylinder having a soft resilient expansible cushioning nose adapted to contact the dynamite and to form an airtight closure for the bore hole, and a relatively rigid rearward portion, said cylinder being formed with spaced bores extending through both the nose and rigid portion for the reception of the ignition wires.

Other objects of the invention will appear as the following description of a preferred and practical embodiment thereof proceeds.

In the drawing which accompanies and forms a part of the following specification and throughout the several figures of which the same characters of reference have been employed to designate identical parts:

Figure 1 is a longitudinal section through a bore hole showing the arrangement of the explosive and of the safety blasting plug;

Figure 2 is a perspective view of the plug per se; and

Figure 3 is a longitudinal section through the safety blasting plug indicating the method of withdrawing it after a misfire.

Referring now in detail to the several figures, the numeral 1 represents a rock mass or mineral vein into which the bore hole 2 has been drilled and in the rear portion of the bore hole are the sticks of dynamite 3. It is customary to insert a detonating cap 4 in the final stick of explosive, the cap being furnished with ignition wires 5 and 6 leading to a source of current and suitable circuit closing mechanism, not shown. In the absence of the safety blasting plug, it is customary to ram the dynamite into the bore hole with a wooden ram. The ignition wires are, of course, in the way of the ram and are pressed into contact with the side of the bore hole, in which position the insulation is apt to be abraded by the reciprocatory movements of the ram so that some portion of the wires is likely to be laid bare and come in contact with the earth or rock. Under these conditions a short circuit will occur at the time of firing so that the charge is not ignited, the result being that when the adjacent charges are exploded, the dead charge will be thrown out into the debris of the explosion, including the stick of dynamite having the detonating cap, so that later when the workmen are digging into the debris, a metallic tool may come into contact with the unfired detonating cap, resulting in a disastrous explosion. Even in the event that the unfired charge remains in place in the bore hole, it can only be refired by withdrawing the final stick having the detonating cap and replacing with a cap having intact ignition wires. Since an auger is employed for withdrawing the final stick of explosive, there is grave danger of the auger striking the detonating cap and exploding the charge.

The safety blasting plug of the present invention comprises a cylindrical body which in general, is designated by the reference character 7 having a forward portion or nose 8 formed of soft rubber or equivalent material, for example, sponge rubber is found to be a satisfactory material. The rearward portion 9 of the blasting plug is of relatively rigid material, of which hard rubber is a characteristic example, to which the nose portion 8 is vulcanized or otherwise secured. It

is one of the features of the invention that no metal is employed in its fabrication, so that there is no danger of a spark being struck up by contact of a tool with any portion of the blasting plug. The unitary structure thus constituted is provided with separate longitudinal bores 10 and 11.

In the use of my safety blasting plug, the sticks of explosive 3 are placed in the bore hole in the usual manner. They may be tamped or not, according to the usual practice, but it is not necessary to ram them into the bore hole.

The final stick of explosive is furnished with the detonating cap in the usual manner, but before inserting it in the bore hole, it is placed in conjunction with the soft end of the blasting plug, the ignition wires 5 and 6 being threaded through the bores 10 and 11. The blasting plug with the final stick of explosive thus connected to it is inserted in the bore hole. A wooden ram is used to push it into place, but on account of the novel construction of the blasting plug, it is not necessary to ram it home by repeated reciprocations of the tamper as in the older method. When the blasting plug together with the final stick of explosive is pushed into place, the soft nose 8 is deformed radially as indicated at 12 in Figure 1, filling the bore hole and conforming to the irregularities in the walls thereof, so as to constitute a gas-tight plug. It thus becomes jammed in place, so that when the tamper is removed, the soft nose of the plug remains in expanded position as shown in Figure 1. The soft rubber or its equivalent at the advance end of the plug affords a cushion for the adjacent stick of explosive, while the bores 10 and 11 keep the ignition wires apart and protect them from injury and short circuiting.

Figure 1 shows that the anterior end of the bore hole has been blocked by a pair of sand bags 13. These are merely pushed into place as required by the mining regulations and while they necessarily force the ignition wires to the side of the bore hole, they do not abrade or otherwise injure the insulation. The presence of these sand bags is unnecessary, since the bore hole is properly occluded by the expanded soft nose portion of the blasting plug.

In the event that there should be some inherent defect in the detonating cap so that the latter does not explode, it is necessary to remove the blasting plug and the final stick of explosive, to replace the defective cap with a fresh one and to explode the charge by a new ignition. Figure 3 shows that the rearward or relatively rigid portion of the blasting plug may be penetrated by an auger 14, by means of which the entire blasting plug may be removed. In the process of removal, there is no risk of the auger coming into contact with the detonating cap. Since the ig-

niton wires are threaded through the bores 10 and 11, and since the final charge of explosive is loose in the bore hole, not having been rammed or tamped, it will come out with the blasting plug. It is merely necessary then to remove the blasting plug, take out the defective cap, insert another one in the stick of dynamite and reintroduce the ignition wires through the bores in the blasting plug.

While I have in the above disclosure described what I believe to be a preferred and practical embodiment of my invention, it is to be understood that the specific details of construction as shown and described are merely by way of example, and that other materials than rubber, but having the same relative characteristics of the front and rearward portions of the blasting plug, may be employed without departing from the spirit and scope of the invention.

What I claim is:

1. Safety blasting plug comprising a cylindrical member having a radially expansible relatively soft anterior portion, and a relatively rigid rearward portion, said member having spaced longitudinal bores extending through both said portions.

2. Safety blasting plug comprising a cylindrical member having a radially expansible relatively soft anterior portion, and a relatively rigid rearward portion, said portions being integrally united, said member having spaced longitudinal bores extending through both said portions.

3. Safety blasting plug comprising an altogether non-metallic cylindrical member having a radially expansible relatively soft anterior portion, and a relatively rigid rearward portion, said portions being integrally united and said member having spaced longitudinal bores extending through both said portions.

4. Safety blasting plug comprising an altogether non-metallic cylindrical member including an anterior portion of soft rubber and a posterior portion of relatively hard rubber, said portions being integrally united, said member having spaced longitudinal bores extending through both said portions.

In testimony whereof I affix my signature.
RICHARD J. HEITZMAN.